

Heart Failure

ORG: M-190 (ISC)

[Link to Codes](#)**MCG Health**Inpatient & Surgical
Care
28th Edition

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Care Planning - Inpatient Admission and Alternatives

Clinical Indications for Admission to Inpatient Care

Note: Some patients may be appropriate for Observation care. For consideration of observation care, see Heart Failure: Observation Care [ISC](#).

Note: For patients with clinically active secondary conditions, see Heart Failure Multiple Condition Management Guidelines.

- Admission is indicated for **1 or more** of the following(1)(2)(3)(4):[N](#)
 - Hemodynamic instability (2)
 - Severe electrolyte abnormalities requiring inpatient care
 - Cardiac arrhythmias of immediate concern
 - Acute myocardial ischemia causing or associated with failure. See Angina [ISC](#) or Myocardial Infarction [ISC](#) as appropriate.
 - Pulmonary edema that is very severe (eg, invasive or noninvasive assisted ventilation needed, imminent or likely, or need for 100% oxygen to keep oxygen saturation above 90%)
 - Anasarca or severe peripheral edema (eg, impairs ability to void or ambulate)(2)(7)
 - ☐ Pulmonary edema that is persistent, as indicated by **ALL** of the following:
 - New need for oxygen therapy to keep oxygen saturation above 90% (or increased FiO₂ need from baseline)
 - Has not improved sufficiently with observation care (eg, appropriately dosed IV diuretics)[A]
 - Tachypnea that persists despite observation care
 - Dyspnea (above baseline) that persists despite observation care
 - Altered mental status that is severe or persistent
 - Increased creatinine with reduction of more than 50% in estimated glomerular filtration rate from baseline[B]
 - [eGFR - Adult Calculator](#)
 - Progressively (ongoing) rising creatinine with reduction of more than 25% in estimated glomerular filtration rate from baseline[B]
 - [eGFR - Adult Calculator](#)

- o Pulmonary artery catheter monitoring needed

Alternatives to Admission

- Alternatives include(1)(2)(3)(4)(11)(12)(13)(14):[N](#)
 - o For arrhythmia: emergency department or rapid treatment site intervention(15):
 - Pharmacologic therapy to slow supraventricular arrhythmias
 - Cardioversion (electrical or pharmacologic) to sinus rhythm
 - o For heart failure with pulmonary congestion or peripheral edema:
 - Outpatient therapy (eg, renin-angiotensin inhibitor (angiotensin receptor-neprilysin inhibitor preferred), diuretics, beta-blocker, aldosterone antagonist, sodium-glucose cotransporter-2 inhibitor (eg, dapagliflozin, empagliflozin), vericiguat, or vasodilator (hydralazine, isosorbide), patient education, salt and fluid restriction)[A](#)[J](#)[C](#)(1)(14)
 - Multidisciplinary team care with home care support(1)(17)
 - Emergency department or rapid treatment site care with oxygen, respiratory therapy, IV diuretics, correction of fluid and electrolyte balance, and control of arrhythmias(11)(18)(19)
 - Recovery facility care for ongoing parenteral medication, oxygen therapy, weight and fluid balance, laboratory testing, electrolyte correction, and management of comorbidities
 - Skilled nursing facility for medication adjustment, weight and fluid balance, nursing assessment, and nutritional support(20)
 - o Observation. See Heart Failure: Observation Care [ISC](#) guideline as appropriate.

Alternative Care Planning

- Care planning needs for patient not requiring admission may include(1)(3)(7)(14)(18)(21)(22):
 - o Diagnostic tests, including(23):
 - Laboratory testing, including CBC, electrolytes, fasting glucose, glycosylated hemoglobin, kidney function, urinalysis, liver function, lipid profile, transferrin, HIV, and thyroid function
 - Cardiac biomarkers (eg, BNP, N-terminal pro-BNP level, and Troponin)(24)(25)
 - Vitamin D level, iron studies(26)
 - Chest x-ray
 - ECG
 - Oxygen saturation
 - Echocardiogram
 - Chest ultrasound(27)
 - Radionuclide ventriculography
 - Cardiac MRI
 - Ventricular viability testing (eg, nuclear study or cardiac MRI, for patients with ischemic cardiomyopathy)(28)
 - Pulmonary function tests
 - Pulmonary CT angiogram or lung ventilation perfusion scan
 - o Treatments and procedures, including:
 - Pharmacologic treatment (eg, diuresis, beta-blockers, renin-angiotensin inhibitors, aldosterone blockers, and sodium-glucose cotransporter-2 inhibitors)[A](#)[J](#)[C](#)(1)(14)(18)(29)
 - Antihypertensive medication^[D]
 - Cardioversion (electrical or pharmacologic) for supraventricular tachycardia or atrial fibrillation(15)
 - o Consultation, assessment, and other services, including:
 - Cardiology consultation
 - Electrophysiology specialist consultation (eg, arrhythmia aspect of acute presentation, resynchronization therapy, cardioverter-defibrillator placement).(1)(30)(31)(32)(33) See Electrophysiologic Study and Intracardiac Catheter Ablation [ISC](#), Left Atrial Appendage Closure, Percutaneous [ISC](#), or Electrophysiologic Study and Implantable Cardioverter-Defibrillator (ICD) Insertion [ISC](#) as appropriate.
 - Smoking cessation counseling(1)(2)
 - Nutrition counseling(1)(16)
 - Cardiac rehabilitation(34)(35)
 - Vaccinations against respiratory illness (eg, influenza, pneumococcal, respiratory syncytial virus, COVID-19)(1)
 - Palliative care(1)(36)
 - o Discharge Planning as appropriate(2)(37)(38)
 - o Patient, family, and caregiver education as appropriate. See Heart Failure: Patient Education for Clinicians [SR](#).(37)(38)

Hospitalization

Optimal Recovery Course

Day	Level of Care	Clinical Status	Activity	Routes	Interventions	Medications
1	- ICU[E] or intermediate care[F] or floor - Social Determinants of Health Assessment - Readmission Risk Assessment - Extended Stay Risk Assessment - Discharge planning	Clinical Indications met[G] - Tachypnea, edema, and dyspnea - Possible hypoxia and hypotension	Initial bed rest	- Parenteral medications - Oral hydration - Oral diet	- ECG, CXR, ABG - Cardiac biomarkers, BNP - Possible pulmonary artery catheter[H] - Oxygen - Possible echocardiogram - Digoxin level if previously prescribed - Electrolytes - Weigh - Possible fluid restriction	- IV diuretics[A] - Renin-angiotensin inhibitor (angiotensin receptor-neprilysin inhibitor preferred)[C] - Beta-blocker[I] - Aldosterone antagonist - Sodium-glucose cotransporter-2 inhibitor[J] - Possible IV vasodilator (eg, nitrates, hydralazine)
2	- Floor - Social Determinants of Health Assessment - Readmission Risk Assessment - Extended Stay Risk Assessment	Hemodynamic stability Mental status at baseline No evidence of myocardial ischemia Cardiac rate and rhythm acceptable Oxygenation at baseline or improved Pulmonary edema absent or improved	Advance activity as tolerated	- Taper parenteral medications - Oral diet - Adjust fluid restriction	Pulmonary catheter absent - Weigh - Possible electrolytes[K] - Possible CXR, ECG, BNP - Oxygen	- Diuretics - Renin-angiotensin inhibitor - Beta-blocker - Aldosterone antagonist - Sodium-glucose cotransporter-2 inhibitor - Possible nitrates, hydralazine
3	- Social Determinants of Health Assessment - Readmission Risk Assessment - Extended Stay Risk Assessment - Floor to discharge - Complete discharge planning	Hemodynamic stability Tachypnea absent Oxygenation at baseline or acceptable for next level of care Dyspnea at baseline or acceptable for next level of care Cardiac rate and rhythm acceptable Pulmonary edema absent or acceptable for next level of care Peripheral or sacral edema absent or	Ambulatory or acceptable for next level of care	Oral hydration[L] Oral medications or regimen acceptable for next level of care Oral diet or acceptable for next level of care	Weigh	- Diuretics - Renin-angiotensin inhibitor - Beta-blocker - Aldosterone antagonist - Sodium-glucose cotransporter-2 inhibitor - Possible nitrates, hydralazine

		acceptable for next level of care • Mental status at baseline • Volume status acceptable on oral medication • Renal function at baseline or acceptable for next level of care • Electrolyte abnormalities absent or acceptable for next level of care • Precipitating factors absent or controlled • Discharge plans and education understood				
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(1)(2)(6)(13)(45)[N](#)

Care Task List

Recovery Milestones are indicated in **bold**.

Goal Length of Stay: 2 days

Note: Goal Length of Stay assumes optimal recovery, decision making, and care. Patients may be discharged to a lower level of care (either later than or sooner than the goal) when it is appropriate for their clinical status and care needs.

Extended Stay

Note: See Heart Failure Multiple Condition Management Benchmarking Table for more detailed information.

Minimal (a few hours to 1 day), Brief (1 to 3 days), Moderate (4 to 7 days), and Prolonged (more than 7 days).

- Extended stay beyond goal length of stay may be needed for(1)(2)(6)(7):
 - Failure to meet discharge criteria (recovery milestones within final day in Optimal Recovery Course)
 - Expect brief stay extension.
 - Acute cardiac ischemia, confirmed or suspected as precipitant or consequence of heart failure(28)
 - Cardiac ischemia may prompt rapid cardiac catheterization and revascularization with angioplasty, CABG, or thrombolytic therapy.
 - Anticipate possible temporary perioperative mechanical cardiac support (eg, intra-aortic balloon pump).
 - Anticipate intervention within 24 hours.
 - Expect brief stay extension.
 - Cardiogenic shock(39)(40)(46)
 - Anticipate rapid supportive therapy and intervention for treatable causes.
 - Pulmonary artery catheter, inotropic agents and vasopressor agents, continuous positive airway pressure or noninvasive ventilation, intra-aortic balloon pump, ventricular assist device, or cardiac catheterization may be used as indicated.(41)(43)
 - Anticipate completion of evaluation and trial of tapering parenteral support medications in 24 to 48 hours.
 - Treat identifiable causes:
 - For acute valvular disease (eg, chordae tendineae rupture, aortic dissection with valve involvement, critical stenosis): rapid cardiac surgery if possible
 - For pericardial tamponade: pericardiocentesis
 - For pulmonary embolus: anticoagulation, possible thrombolytics, possible embolectomy, possible caval filter
 - Expect brief to moderate stay extension.
 - See Hemodynamic Instability: Common Complications and Conditions [ISC guideline](#) as appropriate.
 - Acute kidney injury (stage 2) or Acute renal failure (stage 3 acute kidney injury)

- Anticipate possible dialysis to control fluid overload if diuresis is not sufficient.
 - Expect brief stay extension.
- Stage IV chronic kidney disease (estimated glomerular filtration rate of less than 30 mL/min/1.73m² (0.50 mL/sec/1.73m²), and not previously on chronic dialysis)  eGFR - Adult Calculator(47)(48)(49)
 - Anticipate need for more intensive diuretic therapy (higher dose or second diuretic) and electrolyte monitoring.
 - Ultrafiltration or renal replacement therapy may be required for diuretic resistance or failure.
 - Expect brief stay extension.
- Respiratory failure (eg, need for noninvasive or invasive assisted ventilation)(50)
 - Anticipate aggressive diuresis, respiratory support, and frequent monitoring of clinical status.
 - Expect brief to moderate stay extension.
 - See Respiratory Insufficiency: Common Complications and Conditions  ISC guideline as appropriate.
- Concomitant pneumonia
 - Anticipate clinical testing (eg, imaging, blood tests) and treatment (eg, antibiotics).
 - Expect brief stay extension.
- Significant electrolyte abnormality (severe hyponatremia, hypokalemia)
 - Anticipate fluid restriction, salt or electrolyte repletion.
 - Expect brief stay extension.
- Newly diagnosed (new-onset) atrial fibrillation(15)
 - Anticipate medication to control heart rate, and possible need for cardioversion (electrical or pharmacologic).
 - Patient may need initiation of anticoagulation. See Anticoagulation Requirement: Common Complications and Conditions  ISC.
 - Expect brief stay extension.

See Common Complications and Conditions  ISC for further information.

Hospital Care Planning

- Hospital evaluation and care needs may include(1)(2)(4)(7)(14)(22)(40):
 - Diagnostic test scheduling and completion, including(23)(41):
 - CBC, electrolytes, urinalysis, digoxin level, kidney function, liver function, lipid panel, thyroid function, fasting glucose, coagulation profile, and glycosylated hemoglobin
 - Cardiac biomarkers (eg, BNP, N-terminal pro-BNP level, Troponin)(24)(25)(41)
 - Vitamin D level, iron studies(26)(57)
 - Chest x-ray
 - ABG
 - Echocardiogram
 - Radionuclide ventriculography
 - Cardiac MRI
 - Treadmill stress test
 - Stress echocardiogram or nuclear stress-test imaging(28)
 - Ventricular viability testing (eg, nuclear study or gadolinium-enhanced cardiac MRI, for patients with ischemic cardiomyopathy)(28)
 - Pulmonary function tests
 - Lung ventilation perfusion scan or pulmonary CT angiogram
 - Treatment and procedure scheduling and completion, including:
 - Medication (eg, diuretics, renin-angiotensin inhibitor (angiotensin receptor-neprilysin inhibitor preferred), aldosterone antagonist, beta blockade,^[I] ivabradine, sodium-glucose cotransporter-2 inhibitor (eg, dapagliflozin, empagliflozin, sotagliflozin),^[J] vericiguat, or vasodilator (eg, intravenous nitroglycerin or nitroprusside))^{[A][C]}(1)(14)
 - Antihypertensive medication^[D]
 - Thromboembolic prophylaxis
 - Intravenous iron(14)(57)
 - Thoracentesis
 - Pulmonary artery catheterization^[H]
 - Coronary angiography(28)
 - Endomyocardial biopsy
 - Noninvasive ventilation or continuous positive airway pressure(58)
 - Timely discontinuation of monitoring (eg, pulmonary artery catheter)
 - Timely discontinuation of therapies (eg, switch to oral diuretics)
 - Intravenous inotropic agent (eg, dobutamine, milrinone, for shock, end organ dysfunction, persistent symptoms or for palliative care)(1)(59)
 - Mechanical circulatory support (eg, ventricular assist device)(41)(43)(60)
 - Ultrafiltration for diuresis-resistant fluid overload(61)

- Consultation, assessment, and other services scheduling and completion, including:
 - Cardiology consultation(16)
 - Electrophysiology specialist consultation (eg, arrhythmia aspect of acute presentation, resynchronization therapy, cardioverter-defibrillator placement).(1)(30)(31)(32)(33) See Electrophysiologic Study and Intracardiac Catheter Ablation [ISC](#), Left Atrial Appendage Closure, Percutaneous [ISC](#), or Electrophysiologic Study and Implantable Cardioverter-Defibrillator (ICD) Insertion [ISC](#) as appropriate.
 - Cardiac rehabilitation consultation(34)(35)
 - Smoking cessation counseling(2)
 - Nutrition counseling(1)(16)
 - Transplant team consultation(28)(62)
 - Sleep medicine consultation(16)(63)
 - Vaccinations against respiratory illness (eg, influenza, pneumococcal, COVID-19)(1)
 - Palliative care consultation(1)(16)(36)
- Identification of patient at high risk for readmission to prioritize transition and post-acute care:
 - ☐ Risk of readmission is increased by presence of **1 or more** of the following(37)(64)(65)(66)(67)(68)(69)(70)(71):
 - Hospitalization for heart failure in past year or for any reason in past 6 months(51)(52)(53)(54)(72)
 - Emergency department visit in past 6 months
 - No source of outpatient care other than emergency department (eg, no primary care provider)(73)
 - Severe care transition barriers (eg, no caregiver, homeless)(74)(75)
 - Active substance abuse(56)
 - Skin ulcer present on admission (eg, decubitus ulcer)(55)
 - Renal insufficiency (GFR less than 60 mL/min/1.73m² (1.00 mL/sec/1.73m²))(51)(52)(53)(54)(55)(56)(70)(72)(76)
 -  eGFR - Adult Calculator
 - Moderate or severe aortic stenosis(51)(52)(53)(54)(55)(72)
 - Anemia (hematocrit less than 30% (0.30))(51)(52)(53)(54)(72)
 - Metastatic solid tumor (eg, lung cancer, breast cancer)(70)
 - Advanced liver disease (eg, cirrhosis with portal hypertension, history of variceal bleed)(70)
- Monitoring patient's status for deterioration and comorbid conditions (see Inpatient Monitoring and Assessment Tool [SR](#)); key items include:
 - Complicating illness (eg, COPD, diabetes)(77)
 - Cardiovascular status, including:
 - Cardiac rate and rhythm, blood pressure (including postural changes), and peripheral pulses
 - Activity tolerance
 - Access site and limb surveillance for bleeding, ischemia, and neurologic function if catheterization is performed or a mechanical circulatory support device is placed(41)
 - Oxygen requirements to maintain saturation of arterial oxygen greater than 90%
 - Fluid balance
 - Electrolytes and renal function
 - Severity of peripheral or presacral edema
 - Daily weight
 - Presence of abdominal pain
 - Depression(78)

Discharge

Discharge Planning

- Discharge planning includes[M]:
 - Assessment of needs and planning for care, including(80)(81):
 - Develop and modify treatment plan (involving multiple providers) as needed.
 - Evaluate and address preadmission functioning as needed.
 - Evaluate and address psychosocial status issues as indicated. See Psychosocial Assessment [SR](#) for further information.
 - Evaluate and address social determinants of health (eg, housing, food). See Social Determinants of Health Screening Tool [SR](#) for further information.(79)
 - Evaluate and address patient or caregiver preferences as indicated.
 - Identify skilled services needed at next level of care, with specific attention to:
 - Cardiovascular status assessment(82)
 - Medication management, adherence instruction, and side effects assessment(83)
 - Nutrition and hydration management(84)(85)
 - Ongoing education required(84)

- Psychosocial assessment, management, and referrals(86)
- Early identification of anticipated discharge destination; options include(81)(87):
 - Home, considerations include:
 - Home safety assessment. See Home Safety Assessment [SR](#) for further information.
 - ☐ Patient safe to go home; examples include(88)(89)(90):
 - Medical status stable for patient's condition
 - Functional care can safely be provided with available resources.
 - Mental status stable for patient's condition
 - Medication availability confirmed and reconciliation complete
 - Patient/caregiver education completed with written discharge instructions provided
 - Community resources identified and referrals made, as needed
 - Home care arranged, if indicated
 - Necessary medical equipment delivery arranged or available in home, if indicated
 - Necessary medical supplies ordered, or patient/caregiver can obtain, if indicated
 - Access to follow-up care
 - Self-management ability if appropriate. See Activities of Daily Living (ADL) and Instrumental Activities of Daily Living (IADL) Assessment [SR](#) for further information.
 - Caregiver need, ability, and availability
 - Post-acute skilled care or custodial care as indicated. See Discharge Planning Tool [SR](#) for further information.
- Transitions of care plan complete, including(81)(87)(91):
 - Patient and caregiver education complete. See Heart Failure: Patient Education for Clinicians [SR](#) for further information.
 - See Teach Back Tool [SR](#) for further information.
 - ☐ Medication reconciliation completion includes(83)(92):
 - Compare patient's discharge list of medications (prescribed and over-the-counter) against provider's admission or transfer orders.
 - Assess each medication for correlation to disease state or medical condition.
 - Report medication discrepancies to prescribing provider, attending physician, and primary care provider, and ensure accurate medication order is identified.
 - Provide reconciled medication list to all treating providers.
 - Confirm that patient or caregiver can acquire medication.
 - Educate patient and caregiver.
 - Provide complete medication list to patient and caregiver.
 - Importance of presenting personal medication list to all providers at each care transition, including all provider appointments
 - Reason, dosage, and timing of medication (eg, use "teach-back" techniques)(93)
 - Encourage communication between patient, caregiver, and pharmacy for obtaining prescriptions, setting up home medication delivery, and reviewing for drug-drug interactions.
 - See Medication Reconciliation Tool [SR](#) for further information.
 - Plan communicated to patient, caregiver, and all members of care team, including(94):
 - Inpatient care and service providers
 - Primary care provider
 - All post-discharge care and service providers
 - Appointments planned or scheduled, which may include:
 - Primary care provider(85)
 - Cardiologist(95)
 - Dietitian(85)
 - Follow-up evaluation call. See Post-Acute Care Follow-Up Call Tool [SR](#) for more information.(96)
 - Palliative or end-of-life care provider(97)(98)(99)
 - Rehabilitation therapy services(85)
 - Specialists for management of comorbidities as needed(100)
 - Other
 - Outpatient testing and procedure plans made, which may include:
 - Echocardiogram
 - Exercise stress test(85)
 - Laboratory testing(101)
 - Other
 - Referrals made for assistance or support, which may include:
 - Alcohol and other drug abuse or dependence treatment(102)
 - Community services
 - Educational program (eg, chronic condition management, self-management)(101)(103)

- End-stage disease planning(85)(99)
- Financial, for follow-up care, medication, and transportation
- Self-help or support groups(85)
- Social services (eg, social programs, advance directives)(103)
- Tobacco use treatment(85)(101)(104)
- Other
- Medical equipment and supplies coordinated (ie, delivered or delivery confirmed), which may include(85):
 - Blood pressure cuff
 - Weight scale
 - Other

Discharge Destination

- Post-hospital levels of admission may include:
 - Home.
 - Home healthcare. See Home Care Indications for Admission Section [HC](#) in Heart Failure guideline in Home Care.
 - Recovery facility care. See Recovery Facility Care Indications for Admission Section [RFC](#) in Heart Failure guideline in Recovery Facility Care.

Evidence Summary

Criteria

The evidence for the clinical indications found in this guideline includes 2 published peer reviewed articles, 3 specialty society or other evidence-based guidelines, and 2 book sections.

Analysis of national hospital discharge data for a commercially insured population shows 73.1% of patients seen in an emergency department with a principal diagnosis of heart failure were admitted to inpatient care.(5) **(EG 3)** Analysis of national hospital discharge data for a Medicare-insured population shows 81.0% of patients seen in an emergency department with a principal diagnosis of heart failure were admitted to inpatient care.(5) **(EG 3)** Analysis of an all-payer database shows a mean length of stay of 5.4 days for patients admitted to inpatient care.(6) **(EG 3)** An evidence-based specialty society guideline notes that patients with decompensated heart failure (eg, worsening pulmonary edema with dyspnea, tachypnea, and/or hypoxemia) may require hospitalization with rapid initiation and escalation (if necessary) of intravenous diuretics in addition to other medications recommended to reduce morbidity and mortality in heart failure patients.(1) **(EG 2)** Patients with severe presentations (eg, dangerous arrhythmia, myocardial ischemia, hypotension, severe or persistent altered mental status, acute kidney injury) may require evaluation for intensive care.(1) **(EG 2)** An emergency medicine textbook recommends inpatient care for heart failure patients who present with findings categorizing the patient as at moderate or high risk of short-term mortality and morbidity.(3) **(EG 2)** Examples of these findings include hypotension, need for respiratory assistance (ie, CPAP, BiPAP, mechanical ventilation), dangerous arrhythmia, myocardial ischemia, persistent pulmonary edema leading to hypoxemia that persists despite an adequate trial of intravenous diuretic therapy, new or worsening hyponatremia, severe edema, and acute kidney injury.(3) **(EG 2)**

Alternatives

Analysis of 52,055,378 adult patients seen in the emergency department for heart failure over a 10-year period from the Nationwide Emergency Department Sample database reported 30% were treated without inpatient admission.(12) **(EG 2)** Analysis of 1,305,639 adult patients seen in the emergency department for heart failure from the Nationwide Inpatient Sample database reported 29% were treated without inpatient admission.(13) **(EG 2)** Analysis of 1756 pediatric patients (mean age 4.9 years) seen in the emergency department for heart failure from the Kids' Inpatient Database reported 25% were treated without inpatient admission.(13) **(EG 2)**

Hospitalization

A specialty guideline recommends 4 medication classes as guideline-directed medical therapy (GDMT) for patients with heart failure with reduced ejection fraction (HFrEF): beta-blockers, renin-angiotensin inhibitors (angiotensin receptor-neprilysin inhibitor preferred), aldosterone blockers, and sodium-glucose cotransporter-2 inhibitors.(1) **(EG 2)** This same guideline recommends the following for patients hospitalized with HFrEF absent clear contraindications: pre-existing GDMT should be continued and optimized (titration to therapeutic dose begun as inpatient), GDMT should not be discontinued due to mild decrease in renal function or blood pressure, GDMT should be initiated while hospitalized when clinical stability achieved (if all drug classes not on board), and if GDMT is discontinued it should be reinstated as soon as possible.(1) **(EG 2)** An expert panel also recommends these 4 medication classes as GDMT for heart failure with preserved ejection fraction (HFpEF).(16) **(EG 2)**

A specialty clinical guideline, based upon a systematic review of the evidence, concludes that heart failure patients with evidence of significant fluid overload should be promptly treated with intravenous loop diuretics (eg, initial dose twice the home regimen in terms of oral to parenteral milligram equivalents) and titrated to resolution of clinical signs and symptoms of congestion (eg, serial doubling of dosage as bolus or continuous infusion).(1) **(EG 2)** This guideline further recommends increasing the diuretic dose or adding a second

diuretic (eg, thiazide diuretic or aldosterone antagonist with diuretic properties) until an effective regimen is identified that is adequate to relieve congestion.(1) **(EG 2)**

Readmission risk and reduction: Adjusted analysis of Medicare claims data of 1.1 million patients and the medical records of over 64,000 patients admitted for heart failure found that independent predictors of 30-day readmission include the concurrent comorbidities of renal insufficiency or failure, aortic stenosis, or anemia (hematocrit less than 30% (0.30)).(51) **(EG 2)** A multivariate-adjusted study of 614 heart failure inpatients (mean age 77 years) found that anemia and renal insufficiency were independently associated with higher 30-day hospital readmission rates.(52) **(EG 2)** Examination of a cohort of 1372 patients (mean age 57 years) hospitalized for heart failure found that having been previously admitted secondary to a heart failure exacerbation within the past year was independently associated with a higher rate of 30-day readmission.(53) **(EG 2)** Analysis of a database including over 59,000 patients hospitalized for heart failure found that patient hospitalization or an emergency department visit within the previous 6 months were independent predictors of 30-day readmission.(54) **(EG 2)** A multivariate-adjusted analysis of commercial insurance and Medicaid databases including 4584 patients with systolic heart failure (mean age 55 years) found that independent risk factors for 30-day all-cause readmission rates included pre-admission skin ulcer, chronic renal insufficiency, and valvular heart disease.(55) **(EG 2)** A multivariate-adjusted analysis of an all-payer database reported that independent risk factors for 30-day all-cause readmission rates for patients hospitalized with heart failure included active substance abuse and chronic renal insufficiency.(56) **(EG 2)**

Length of Stay

Analysis of 950,045 adult patients admitted with heart failure from the Nationwide Inpatient Sample database found that 25% of patients were discharged in 2.1 days or less.(13) **(EG 2)** A study including 978,135 patients (mean age 72 years) hospitalized with a primary diagnosis of heart failure found that 25% of patients were discharged within 2 days.(45) **(EG 2)** Analysis of national hospital discharge data shows 23% of hospitalized patients with the primary diagnosis of heart failure discharged in 2 days or less.(6) **(EG 3)**

Rationale

Use of this MCG care guideline helps the clinician identify patient-specific complex clinical factors that make it reasonable to expect a necessity for hospital care across 2 or more midnights. The evidence-based clinical indication criteria assist the clinician in the decision to appropriately admit a patient to inpatient care. For Medicare enrollees, MCG care guidelines also support the clinician in the decision to appropriately admit a patient to inpatient care when there is not an expectation of 2 or more midnights of hospital care (eg, CMS' Two-Midnight Rule exceptions).

Use of these evidence-based clinical criteria to support decision making around the need for inpatient treatment is of clinical benefit to the patient and is crucial for quality patient care. Use of evidence-based criteria ensures patients receive the most appropriate care for their specific condition, reduces unnecessary risks, promotes recovery, and provides a standard that can reduce disparities in care. Not only do evidence-based criteria help align treatment decisions with best practice, but they can also help reduce unwarranted variation in care by fostering consistency and equality in healthcare provision across regions and facilities. Appropriate use of the evidence-based clinical criteria in conjunction with the Supplemental Medicare Criteria ensures Medicare patients receive full access to Medicare benefits (eg, inpatient care), without risk of delay or decreased access, based on variables such as clinical severity of illness, length of provision of medically necessary hospital-level care, or satisfaction of specified Medicare criteria as directed by CMS. Use of these criteria will help ensure that the patient receives necessary care in the appropriate setting.

Related CMS Coverage Guidance

This guideline supplements but does not replace, modify, or supersede existing Medicare regulations or applicable National Coverage Determinations (NCDs) or Local Coverage Determinations (LCDs).

Code of Federal Regulations (CFR): 42 CFR 412.3(10); 42 CFR 419.22(105); 42 CFR 422.101(106)

Internet-Only Manual (IOM) Citations: CMS IOM Publication 100-02, Medicare Benefit Policy Manual, Chapter 1 - Inpatient Hospital Services Covered Under Part A(107); CMS IOM Publication 100-02, Medicare Benefit Policy Manual, Chapter 6 - Hospital Services Covered Under Part B(108); CMS IOM Publication 100-02, Medicare Benefit Policy Manual, Chapter 15 - Covered Medical and Other Health Services(109); CMS IOM Publication 100-08, Medicare Program Integrity Manual, Chapter 6, Section 6.5 - Medical Review of Inpatient Hospital Claims for Part A Payment(110)

Medicare Coverage Determinations: Medicare Coverage Database(111)

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Footnotes

[A] A specialty clinical guideline, based upon a systematic review of the evidence, concludes that heart failure patients with evidence of significant fluid overload should be promptly treated with intravenous loop diuretics (eg, initial dose twice the home regimen in terms of oral to parenteral milligram equivalents) and titrated to resolution of clinical signs and symptoms of congestion (eg, serial doubling of dosage as bolus or continuous infusion).(1) This guideline further recommends increasing the diuretic dose or adding a second diuretic (eg, thiazide diuretic or aldosterone antagonist with diuretic properties) until an effective regimen is identified that is adequate to relieve congestion.(1) [A in Context Link 1, 2, 3, 4, 5]

[B] The most accurate estimate of the glomerular filtration rate (GFR) requires a steady state (stable) serum creatinine.(8)(9) However, the clinical setting of acute decompensated heart failure usually does not allow for delayed estimation (ie, waiting for a stable creatinine level). Further, in most instances estimation in the setting of a rising creatinine will underestimate the true decline in GFR.(8) Accepted definitions of acute kidney injury speak to an acute change (rise) in serum creatinine, using these changes as part of the process of defining the stage (severity) of kidney injury.(9) For the purposes of level of care determination, the point is that the degree of heart failure is such that acute kidney injury has occurred or is impending, and therefore inpatient care and monitoring is appropriate. [B in Context Link 1, 2]

[C] A specialty guideline recommends 4 medication classes as guideline-directed medical therapy (GDMT) for patients with heart failure with reduced ejection fraction (HFrEF): beta-blockers, renin-angiotensin inhibitors (angiotensin receptor-neprilysin inhibitor preferred), aldosterone blockers, and sodium-glucose cotransporter-2 inhibitors.(1) This same guideline recommends the following for patients hospitalized with HFrEF absent clear contraindications: pre-existing GDMT should be continued and optimized (titration to therapeutic dose begun as inpatient), GDMT should not be discontinued due to mild decrease in renal function or blood pressure, GDMT should be initiated while hospitalized when clinical stability achieved (if all drug classes not on board), and if GDMT is discontinued it should be reinstated as soon as possible.(1) Vericiguat further reduces the risk of adverse cardiac outcomes in heart failure and may either be started with the standard quadruple therapy outlined above for severe worsening heart failure, used as a substitute agent for patients with contraindication or intolerance to one or more standard medications, or used as an add-on therapy for inadequate response to optimal quadruple therapy.(14) An expert specialty society panel also recommends these 4 medication classes as GDMT for heart failure with preserved ejection fraction (HFpEF).(16) [C in Context Link 1, 2, 3, 4]

[D] Additional antihypertensive agents beyond the recommended heart failure guideline-directed medical therapy (GDMT) medications may be needed for patients with heart failure with preserved ejection fraction (HFpEF) to attain a goal systolic blood pressure less than 130 mm Hg.(16) [D in Context Link 1, 2]

[E] ICU care is indicated for acute respiratory failure, cardiogenic shock, severe arrhythmias, evidence of cardiac ischemia, or need for pulmonary artery catheter, intravenous inotrope or vasopressor, or mechanical circulatory support.(1)(2)(39)(40)(41) See Intensive Care Guidelines [ISC](#) for further information. [E in Context Link 1]

[F] See Intermediate Care Guidelines [ISC](#) for further information. [F in Context Link 1]

[G] See Clinical Indications for Admission to Inpatient Care in this guideline. [G in Context Link 1]

[H] Pulmonary artery catheterization (right-sided heart catheterization) may be needed when diagnosis is unclear (eg, suspected heart failure with left ventricular ejection fraction greater than 40%), for patients with ongoing symptoms or uncertain fluid status following

initial diuretic treatment, or to guide device selection and utilization of mechanical circulatory support.(1)(41)(42)(43) [H in Context Link 1, 2]

[I] Beta-blockers may not be appropriate in the setting of acute pulmonary edema or while intravenous inotropic agents are required.(1) Recommended beta-blockers include carvedilol, bisoprolol, and extended-release metoprolol.(1) [I in Context Link 1, 2]

[J] Specialty guidelines recommend sodium-glucose cotransporter-2 inhibitors for nondiabetic and type 2 diabetic patients with heart failure because they reduce the risk of worsening heart failure, major adverse cardiac events, and cardiac mortality.(1)(44) However, they should be avoided in severe illness, during prolonged fasting, or for surgical procedures.(44) They are not currently FDA approved for patients with type 1 diabetes due to risk of ketoacidosis.(1) [J in Context Link 1, 2]

[K] Electrolyte monitoring may be indicated if patient is actively diuresing.(1) [K in Context Link 1]

[L] Some patients may have their hydration needs met via alternative means (eg, percutaneous endoscopic gastrostomy tube). [L in Context Link 1]

[M] Discharge instructions should be given in the patient's and caregiver's native language using trained language interpreters whenever possible.(79) [M in Context Link 1]

Definitions

Acute kidney injury (stage 2)

- Acute kidney injury (stage 2) with significant clinical findings, as indicated by **ALL** of the following(1)(2)(3):
 - Acute[A] kidney injury (stage 2), as indicated by **1 or more** of the following[B]:
 - Within past 7 days, rise in serum creatinine between 2 and 2.9 times its baseline value (increase of 3 times baseline would qualify as stage 3 acute kidney injury)
 - Decreased urine output,[C] as indicated by **ALL** of the following:
 - Assessment of urine output appropriate (eg, adequate volume status, no urinary obstruction)
 - Urine output less than 0.5 mL/kg/hr for 12 hours
 - Significant clinical findings, as indicated by **1 or more** of the following[B]:
 - Hemodynamic instability
 - Worsening clinical status despite observation care (eg, rising creatinine or urine output remains less than 0.5 mL/kg/hr)
 - Altered mental status
 - Cardiac arrhythmias of immediate concern
 - Severe hypertension (SBP greater than 180 mm Hg or DBP greater than 120 mm Hg in adults or greater than the 95th percentile for age, gender, and height plus 30 mm Hg in pediatric patients)(4)(5)
 - Significant respiratory findings (eg, pulmonary edema) that are severe (eg, Hypoxemia) or persist despite observation care
 - Clinically significant electrolyte abnormality that is severe (eg, hyperkalemia with severe ECG findings)[D] or persists despite observation care
 - Clinically significant metabolic abnormality (eg, metabolic acidosis) that is severe or persists despite observation care

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Footnotes

- A. A specialty society practice guideline defines acute kidney injury as a change in renal function that is known or presumed to have occurred over the previous 7 days.(1)
- B. In determining the stage of acute kidney injury (eg, stage 2 vs stage 3), patients should be staged according to the criteria that give them the highest stage.(1)
- C. The use of urine output as an indicator of acute kidney injury is important, as change in urine output can occur before a noticeable rise in creatinine.(1)(3) It is also suggested that urine output can identify patients with acute kidney injury that may otherwise be missed.(1)(3) However, urine output as a metric is only valid under the appropriate conditions. For example, the patient must not be hypovolemic, there must not be urinary obstruction, and the collection and measurement of urinary output must be complete and accurate.(1)(3) Use of urinary output to define acute kidney injury can be inaccurate in obese patients due to the nonlinear relationship between body weight and urine output. For example, in a 100-kg (220-lb) patient, urine output of 45 mL/hour over 12 hours (adequate output) would qualify as stage 2 acute kidney injury.(1)(3)
- D. ECG findings include peaked T-waves, PR interval greater than 0.20 seconds, QRS interval greater than 0.12 seconds, and arrhythmia.(6)(7)

Acute renal failure (stage 3 acute kidney injury)

- Acute^[A] kidney injury (stage 3),^[B] as indicated by **1 or more** of the following^[C](1)(2)(3):
 - Within past 7 days, rise in serum creatinine to 3 times its baseline value or higher
 - Within past 7 days, rise in serum creatinine to at least 1.5 times its baseline value (increase of 50%) and serum creatinine is greater than 4 mg/dL (354 micromoles/L)
 - In child younger than 18 years, estimated glomerular filtration rate less than 35 mL/min/1.73m² (0.59 mL/sec/1.73m²)
 -  eGFR - Pediatric Calculator and **1 or more** of the following:
 - Within past 7 days, rise in serum creatinine to at least 1.5 times its baseline value (increase of 50%)
 - Within past 48 hours, rise in serum creatinine greater than 0.3 mg/dL (26.5 micromoles/L)
 - Decreased urine output,^[D] as indicated by **ALL** of the following:
 - Assessment of urine output appropriate (eg, adequate volume status, no urinary obstruction)
 - Oligoanuria, as indicated by **1 or more** of the following:
 - Urine output less than 0.3 mL/kg/hour for 24 hours
 - Anuria (urine output less than 0.1 mL/kg/hour) for 12 hours
 - Initiation of renal replacement therapy (RRT) required(4)

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Footnotes

- A. A specialty society practice guideline defines acute kidney injury as a change in renal function that is known or presumed to have occurred over the previous 7 days.(1)
- B. For purposes of these criteria, the term acute renal failure can be viewed as stage 3 acute kidney injury in the KDIGO classification system.(1) Of note, in the ICD-10 diagnostic coding system (main coding scheme used in labeling diagnoses), the relevant codes covering the clinical entity of significant acute kidney injury use the term acute kidney failure (eg, code N17.9).
- C. In determining the stage of acute kidney injury (eg, stage 2 vs stage 3), patients should be staged according to the criteria that give them the highest stage.(1)
- D. The use of urine output as an indicator of acute kidney injury is important, as change in urine output can occur before a noticeable rise in creatinine.(1)(3) It is also suggested that urine output can identify patients with acute kidney injury that may otherwise be missed.(1)(3) However, urine output as a metric is only valid under the appropriate conditions. For example, the patient must not be hypovolemic, there must not be urinary obstruction, and the collection and measurement of urinary output must be complete and accurate.(1)(3) Use of urinary output to define acute kidney injury can be inaccurate in obese patients due to the nonlinear relationship between body weight and urine output. For example, in a 100-kg (220-lb) patient, urine output of 45 mL/hour over 12 hours (adequate output) would qualify as stage 2 acute kidney injury.(1)(3)

Altered mental status

- Altered mental status (ie, different from baseline), as indicated by **1 or more** of the following(1)(2)(3)(4):

- Confusional state (eg, disorientation, difficulty following commands, deficit in attention)
- Lethargy (eg, awake or arousable, but with drowsiness; reduced awareness of self and environment)
- Obtundation (ie, arousable only with strong stimuli, lessened interest in environment, slowed responses to stimulation)
- Stupor (ie, may be arousable but patient does not return to normal baseline level of awareness)
- Coma (ie, not arousable)

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Altered mental status that is severe or persistent

- Altered mental status (ie, different from baseline) that is severe or persistent, as indicated by **1 or more** of the following(1)(2)(3)(4):
 - Confusional state (eg, disorientation, difficulty following commands, deficit in attention) that persists (eg, for more than few hours) despite appropriate treatment (eg, of underlying cause)
 - Lethargy (eg, awake or arousable, but with drowsiness; reduced awareness of self and environment) that persists (eg, for more than few hours) despite appropriate treatment (eg, of underlying cause)
 - Obtundation (ie, arousable only with strong stimuli, lessened interest in environment, slowed responses to stimulation)
 - Stupor (ie, may be arousable but patient does not return to normal baseline level of awareness)
 - Coma (ie, not arousable)

References

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Anasarca

- Anasarca is generalized edema that is widespread and may be more pronounced in dependent areas (eg, back while lying down, legs while sitting up).(1)

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Cardiac arrhythmias of immediate concern

- Cardiac arrhythmias or findings of immediate concern, as indicated by **1 or more** of the following(1)(2)(3):
 - Heart rhythms that are inherently dangerous or unstable, as indicated by **1 or more** of the following(4)(5)(6):
 - Resuscitated ventricular fibrillation or cardiac arrest
 - Ventricular escape rhythm
 - Sustained ventricular tachycardia (30 seconds or more of ventricular rhythm at greater than 100 beats per minute)
 - Nonsustained ventricular tachycardia and **1 or more** of the following:
 - Suspected cardiac ischemia as cause or consequence of ventricular tachycardia
 - Acute myocarditis
 - Unstable cardiac conduction defects, as indicated by **1 or more** of the following(6)(7)(8):
 - Type II second-degree atrioventricular block
 - Third-degree atrioventricular block
 - New-onset left bundle branch block with suspected myocardial ischemia
 - Any heart rhythm and **1 or more** of the following(4)(5)(9)(10)(11):
 - Continuous long-term ECG monitoring needed (eg, initiation of drug requiring monitoring beyond observation care)

- Patient has automatic implanted cardioverter-defibrillator that is repeatedly firing, malfunctioning, or in need of immediate adjustment of settings beyond scope of observation care.
- Heart rhythms of concern due to **1 or more** of the following:
 - Hypotension
 - Respiratory distress
 - Association with other significant symptoms (eg, syncope)(9)(10)(12)

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Care Task List

- Heart failure care tasks to consider as soon as feasible may include:
 - Review results and care implications as appropriate:
 - Cardiac biomarkers (eg, troponin)
 - BNP
 - Repeat ECG
 - Electrolytes
 - Other ordered investigations
 - Schedule, facilitate, or confirm not needed, as appropriate:
 - Echocardiogram
 - Stress test
 - Cardiac angiography
 - Nuclear myocardial scan
 - Cardiology consultation
 - Treatment and comorbidity investigation
 - Thoracentesis
 - Lung ventilation perfusion scan
 - Ultrafiltration
 - Pulmonary function tests
 - Resynchronization therapy (eg, pacemaker or defibrillator placement)
 - Remove pulmonary artery catheter if present, or update treatment plan to achieve this goal.
 - Advance toward planned diet.
 - Test saturation off oxygen and discontinue as appropriate.

- Remove urinary catheter as soon as feasible.
- Advance activity:
 - Out of bed
 - Ambulate short distance, with assistance as needed.
 - Progress to independent activity as tolerated.
 - Deficit assessment by physical therapist, identifying any discharge support requirements
- Order as appropriate:
 - IV medication discontinuation, or adjustment and tapering instructions
 - ACE/ARB oral dosage, or document as not needed
 - Beta-blocker oral dosage, or document as not needed
 - Aldosterone receptor antagonist
 - Outpatient diuretic regimen
 - Conversion of all medications to oral route (or other outpatient regimen)
- Confirm patient status:
 - Volume status is stable on oral intake.
 - Vital signs, mental status, and respiratory function acceptable and stable or improving
 - Ambulatory or at acceptable activity for next level of care without significant dyspnea

Clinically active asthma

- Clinically active asthma refers to asthma necessitating treatment while hospitalized with oral or parenteral (IV) corticosteroids.

Clinically active atrial fibrillation

- Clinically active atrial fibrillation refers to atrial fibrillation necessitating electrical cardioversion or treatment while hospitalized with parenteral (IV) medication to control heart rate (eg, digoxin, diltiazem, metoprolol) or to cardiovert (eg, amiodarone, procainamide).

Clinically active chronic obstructive pulmonary disease

- Clinically active chronic obstructive pulmonary disease (COPD) refers to COPD necessitating assisted ventilation (eg, BPAP, CPAP) or treatment while hospitalized with oral or parenteral (IV) corticosteroids.

Clinically active diabetes

- Clinically active diabetes refers to diabetes necessitating treatment with insulin while hospitalized.

Clinically significant dementia

- Clinically significant dementia refers to dementia necessitating treatment while hospitalized with antipsychotic medication (eg, haloperidol, risperidone) or dementia-specific medication (eg, donepezil, memantine).

Clinically significant malnutrition

- Clinically significant malnutrition refers to patients presenting with findings indicative of poor nutritional status; examples include unintentional weight loss, BMI less than 18.5, inadequate nutritional intake, loss of subcutaneous fat (eg, orbital, triceps, over ribs), or loss of muscle mass (eg, temples, clavicles, hands).  BMI Calculator

Extended Stay Risk Assessment

- Risk of extended length of stay is increased by presence of **1 or more** of the following(1):
 - Acute renal failure (stage 3 acute kidney injury)
 - Significant new-onset electrolyte or metabolic disorder (eg, hyponatremia, hypo-osmolality)
 - Need for mechanical ventilation (eg, acute respiratory failure)

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Heart Failure Multiple Condition Management Guidelines

- For patients who also have:
 - Clinically active asthma, see Heart Failure with Clinically Active Asthma MCM
 - Clinically active atrial fibrillation, see Heart Failure with Clinically Active Atrial Fibrillation MCM
 - Clinically active chronic obstructive pulmonary disease and Clinically active diabetes, see Heart Failure with Clinically Active Chronic Obstructive Pulmonary Disease and Clinically Active Diabetes MCM
 - Clinically active chronic obstructive pulmonary disease, see Heart Failure with Clinically Active Chronic Obstructive Pulmonary Disease MCM
 - Clinically active diabetes, see Heart Failure with Clinically Active Diabetes MCM
 - Clinically significant dementia, see Heart Failure with Clinically Significant Dementia MCM
 - Clinically significant malnutrition, see Heart Failure with Clinically Significant Malnutrition MCM
 - Severe renal failure, see Heart Failure with Severe Renal Failure MCM

Hemodynamic instability

- Hemodynamic instability, as indicated by **1 or more** of the following(1)(2)(3)(4)(5)(6):
 - Vital sign abnormality not readily corrected by appropriate treatment, as indicated by **1 or more** of the following[A]:
 - Tachycardia that persists despite appropriate treatment (eg, volume repletion, treatment of pain, treatment of underlying cause)
 - Hypotension that persists despite appropriate treatment (eg, volume repletion, treatment of underlying cause)
 - Orthostatic hypotension that persists despite appropriate treatment (eg, volume repletion)
 - Hypotension that is severe, as indicated by **ALL** of the following:
 - Hypotension
 - Severe hypotension, as indicated by **1 or more** of the following:
 - Lactate of 2.0 mmol/L (18 mg/dL) or more secondary to hypotension (ie, hypoperfusion)[B](3)(6)
 - Metabolic acidosis (arterial or venous[C] pH less than 7.35) not otherwise explained
 - Mean arterial pressure[A][D] less than 65 mm Hg
 - IV inotropic or vasopressor medication required to maintain adequate blood pressure or perfusion

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Footnotes

- A. Criteria based upon clinician-acquired numeric values (eg, vital signs, oxygen saturation) should be used if they are accurate reflections of the patient's condition. Transitory findings (eg, abnormal only upon initial emergency department intake or only one time out of multiple readings) that rapidly improve with no or minimal treatment usually do not reflect disease severity or risk for deterioration. This does not imply that an initial or one-time reading cannot ever be applicable. The goal is to separate erroneous or incidental findings from those that truly represent the patient's clinical picture.
- B. There are numerous causes of an elevated lactate level. The most common are cardiogenic or hypovolemic shock, severe heart failure, severe trauma, or sepsis. However, there are also other etiologies to consider such as vigorous exercise, seizures, liver disease, or medication use (eg, metformin, beta2-agonists); therefore, interpretation of elevated lactate levels requires consideration of the clinical context (eg, well-appearing post exercise vs hypotensive). The severity and persistence of the elevation can

sometimes be helpful in this differentiation. In most instances of lactic acidosis, blood pH is less than 7.35, with a serum bicarbonate level 20 mEq/L (mmol/L) or lower. However, a coexisting respiratory or metabolic alkalosis can mask these findings.(6)(7)(8)(9)(10)

- C. Analyses have concluded that venous and arterial pH measurements are highly correlated, with small differences between them that are usually not clinically significant.(11)(12)(13)(14)(15) If a mixed acid-base disorder is suspected, an arterial blood gas is indicated. (12)
- D. The mean arterial pressure takes into account both systolic and diastolic blood pressure readings and is calculated as mean arterial pressure (MAP) equals $1/3 \text{ SBP} + 2/3 \text{ DBP}$.(1)(4) A calculator is available at <https://www.mdcalc.com/mean-arterial-pressure-map>.

Hemodynamic stability

- Hemodynamic stability, as indicated by **1 or more** of the following:
 - Hemodynamic abnormalities at baseline or acceptable for next level of care
 - Patient hemodynamically stable, as indicated by **ALL** of the following(1)(2)(3)(4)(5):
 - Tachycardia absent
 - Hypotension absent
 - No evidence of inadequate perfusion (eg, no myocardial ischemia)
 - No other hemodynamic abnormalities (eg, no Orthostatic hypotension)

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Hypotension

- Hypotension, as indicated by **ALL** of the following(1)(2)(3)(4):
 - Not patient baseline (eg, healthy adult with low SBP) or intentional therapeutic goal (eg, low SBP as treatment goal in heart failure)
 - Low blood pressure, as indicated by **1 or more** of the following:
 - SBP less than 90 mm Hg in adult or child 10 years or older[A]
 - Mean arterial pressure[A][B] less than 70 mm Hg in adult or child 10 years or older
 - Decrease in baseline mean arterial pressure of 25% or more,[A][B] with significant signs or symptoms due to lower blood pressure (eg, near syncope, syncope, chest pain)
 - SBP less than sum of 70 mm Hg plus twice patient's age in years in child 1 to 9 years of age[A]
 - SBP less than 70 mm Hg in infant 1 to 11 months of age[A]

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Footnotes

- A. Criteria based upon clinician acquired numeric values (eg, vital signs, oxygen saturation) should be used if they are accurate reflections of the patient's condition. Transitory findings (eg, abnormal only upon initial emergency department intake or only one time out of multiple readings) that rapidly improve with no or minimal treatment usually do not reflect disease severity or risk for deterioration. This does not imply that an initial or one-time reading cannot ever be applicable. The goal is to separate erroneous or incidental findings from those that truly represent the patient's clinical picture.

- B. The mean arterial pressure takes into account both systolic and diastolic blood pressure readings and is calculated as Mean Arterial Pressure (MAP) = $1/3 \text{ SBP} + 2/3 \text{ DBP}$. A calculator is available at <https://www.mdcalc.com/mean-arterial-pressure-map>.

Hypotension absent

- Hypotension absent^[A], as indicated by **1 or more** of the following⁽¹⁾⁽²⁾⁽³⁾⁽⁴⁾:
 - SBP greater than or equal to 90 mm Hg in adult or child 10 years or older
 - Mean arterial pressure^[B] greater than or equal to 70 mm Hg in adult or child 10 years or older
 - Mean arterial pressure^[B] at patient's baseline (eg, healthy adult with low SBP), at intentional therapeutic goal (eg, patient with heart failure), or acceptable for next level of care (eg, blood pressure stable and no significant signs or symptoms due to low blood pressure)
 - SBP greater than or equal to sum of 70 mm Hg plus twice patient's age in years in child 1 to 9 years of age
 - SBP greater than or equal to 70 mm Hg in infant 1 to 11 months of age

References

- Jones D, Di Francesco L. Hypotension. In: McKean SC, Ross JJ, Dressler DD, Scheurer DB, editors. Principles and Practice of Hospital Medicine. 2nd ed. New York, NY: McGraw-Hill Education; 2017:657-64.
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Footnotes

- A. Criteria based upon clinician-acquired numeric values (eg, vital signs, oxygen saturation) should be used if they are accurate reflections of the patient's condition. Transitory findings (eg, abnormal only upon initial emergency department intake or only one time out of multiple readings) that rapidly improve with no or minimal treatment usually do not reflect disease severity or risk for deterioration. This does not imply that an initial or one-time reading cannot ever be applicable. The goal is to separate erroneous or incidental findings from those that truly represent the patient's clinical picture.
- B. The mean arterial pressure takes into account both systolic and diastolic blood pressure readings and is calculated as Mean Arterial Pressure (MAP) = $1/3 \text{ SBP} + 2/3 \text{ DBP}$.

Hypoxemia

- Hypoxemia, as indicated by **1 or more** of the following⁽¹⁾:
 - Patient without baseline need for supplemental oxygen with oxygen saturation (SaO₂) of less than 90% or arterial blood gas partial pressure of oxygen (PO₂) of less than 60 mm Hg (8.0 kPa) on room air^{[A][B]}
 - Patient without baseline need for supplemental oxygen who now requires supplemental oxygen to keep SaO₂ greater than 89% or PO₂ greater than 59 mm Hg (7.9 kPa)^[A]
 - Patient with baseline need for supplemental oxygen who now requires increased supplemental oxygen to maintain oxygenation at baseline or acceptable level

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Footnotes

- A. Pulse oximetry (SpO₂) may underestimate arterial saturation (SaO₂), especially in people with dark skin pigmentation, thick skin, cold body temperature, or who are wearing colored nail polish, and thus may not accurately detect hypoxemia.⁽²⁾ Where appropriate, pulse oximetry should be measured after warming fingers or removing nail polish. Pulse oximetry results should be assessed within the context of the patient's clinical presentation, and performance of an arterial blood gas considered for a more accurate assessment of oxygenation if indicated.⁽²⁾ Specific target values may require adjustment when care is delivered at high altitude.⁽¹⁾⁽³⁾

- B. Criteria based upon clinician-acquired numeric values (eg, vital signs, oxygen saturation) should be used if they are accurate reflections of the patient's condition. Transitory findings (eg, abnormal only upon initial emergency department intake or only one time out of multiple readings) that rapidly improve with no or minimal treatment usually do not reflect disease severity or risk for deterioration. This does not imply that an initial or one-time reading cannot ever be applicable. The goal is to separate erroneous or incidental findings from those that truly represent the patient's clinical picture.

Observation care

- When a Clinical Indication says "...despite (or beyond) observation care," the intent is to signify that, despite an attempt to address a sign, symptom, finding, or treatment need in observation care, a continued need for hospital-based care persists beyond the observation care time frame (ie, Observation Care Discharge Criteria are not met). The time frame of observation care is determined by regulation (eg, CMS's Two-Midnight Rule) or payer-provider contractual agreement. MCG's understanding of observation care is that it must be finite and of short duration. The Two-Midnight Rule for Medicare patients (inpatient is defined as medical necessity for hospital-based care across 2 or more midnights) provides a good example of what is meant by a short duration (ie, observation care should not continue beyond the second midnight).

Orthostatic hypotension

- Orthostatic hypotension,^{[A][B]} as indicated by **1 or more** of the following⁽¹⁾⁽²⁾⁽³⁾:
 - Fall in SBP of 20 mm Hg or more 1 to 3 minutes after patient sits or stands from recumbent position
 - Fall in DBP of 10 mm Hg or more 1 to 3 minutes after patient sits or stands from recumbent position

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Footnotes

- A. Concomitant measurements of the heart rate are important to measure to help diagnose subtypes of orthostatic hypotension (eg, the lack of a compensatory increase in heart rate is typical of autonomic failure and an exaggerated tachycardia may be reflective of volume depletion). However, the heart rate is not a component of the definition of orthostatic hypotension which relies upon blood pressure alone.⁽¹⁾⁽²⁾⁽³⁾
- B. Criteria based upon clinician acquired numeric values (eg, vital signs, oxygen saturation) should be used if they are accurate reflections of the patient's condition. Transitory findings (eg, abnormal only upon initial emergency department intake or only one time out of multiple readings) that rapidly improve with no or minimal treatment usually do not reflect disease severity or risk for deterioration. This does not imply that an initial or one-time reading cannot ever be applicable. The goal is to separate erroneous or incidental findings from those that truly represent the patient's clinical picture.

Readmission Risk Assessment

- Risk of readmission is increased by presence of **1 or more** of the following⁽¹⁾⁽²⁾⁽³⁾⁽⁴⁾⁽⁵⁾⁽⁶⁾⁽⁷⁾⁽⁸⁾⁽⁹⁾:
 - Hospitalization for heart failure in past year or for any reason in past 6 months⁽¹⁰⁾⁽¹¹⁾⁽¹²⁾⁽¹³⁾⁽¹⁴⁾
 - Emergency department visit in past 6 months
 - No source of outpatient care other than emergency department (eg, no primary care provider)⁽¹⁵⁾
 - Severe care transition barriers (eg, no caregiver, homeless)⁽¹⁶⁾⁽¹⁷⁾
 - Active substance abuse⁽¹⁸⁾
 - Skin ulcer present on admission (eg, decubitus ulcer)⁽¹⁹⁾
 - Renal insufficiency (GFR less than 60 mL/min/1.73m² (1.00 mL/sec/1.73m²))⁽⁷⁾⁽¹⁰⁾⁽¹¹⁾⁽¹²⁾⁽¹³⁾⁽¹⁴⁾⁽¹⁸⁾⁽¹⁹⁾⁽²⁰⁾
 -  eGFR - Adult Calculator
 - Moderate or severe aortic stenosis⁽¹⁰⁾⁽¹¹⁾⁽¹²⁾⁽¹³⁾⁽¹⁴⁾⁽¹⁹⁾
 - Anemia (hematocrit less than 30% (0.30))⁽¹⁰⁾⁽¹¹⁾⁽¹²⁾⁽¹³⁾⁽¹⁴⁾
 - Metastatic solid tumor (eg, lung cancer, breast cancer)⁽⁷⁾
 - Advanced liver disease (eg, cirrhosis with portal hypertension, history of variceal bleed)⁽⁷⁾

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Respiratory distress

- Respiratory distress, as indicated by **ALL** of the following(1)(2):
 - Patient with **1 or more** of the following:
 - Dyspnea (difficulty breathing)
 - Tachypnea
 - Abnormal breathing pattern (eg, chest retractions)
 - Other evidence of difficulty breathing
 - Evidence of respiratory compromise, as indicated by **1 or more** of the following:
 - Hypoxemia
 - Altered mental status
 - Other evidence of respiratory compromise (eg, respiratory acidosis)

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Severe electrolyte abnormalities requiring inpatient care

- Severe electrolyte abnormalities requiring inpatient care, as indicated by **ALL** of the following(1)(2)(3)(4):
 - Electrolyte abnormality is not at acceptable patient baseline (eg, treatment effect).
 - Severe abnormality, as indicated by **1 or more** of the following:
 - Sodium[A] less than 130 mEq/L (mmol/L) not corrected to near normal or near chronic baseline despite observation care treatment(5)(6)
 - Sodium[A] less than 135 mEq/L (mmol/L) with severe finding requiring inpatient management (eg, Altered mental status, seizures)(5)(6)
 - Sodium greater than 150 mEq/L (mmol/L) not corrected to near normal or near chronic baseline despite observation care treatment(5)
 - Sodium greater than 145 mEq/L (mmol/L) with severe finding requiring inpatient management (eg, Altered mental status, seizures)(5)
 - Potassium less than 2.5 mEq/L (mmol/L) not corrected to near normal or near chronic baseline despite observation care treatment(7)
 - Potassium less than 3 mEq/L (mmol/L) with severe finding requiring inpatient management (eg, paresis, paralysis, arrhythmia, cardiac conduction disturbance)(7)
 - Potassium greater than 6.5 mEq/L (mmol/L) not corrected to near normal or near chronic baseline despite observation care treatment(7)(8)
 - Potassium greater than 5 mEq/L (mmol/L) with severe finding (eg, ECG findings[B] paresis, paralysis, arrhythmia, cardiac conduction disturbance)(7)(8)
 - Calcium[C] less than 7 mg/dL (1.75 mmol/L) or ionized calcium less than 3.2 mg/dL (0.8 mmol/L) not corrected to near normal or near chronic baseline despite observation care treatment(10)
 - Calcium[C] less than 8.6 mg/dL (2.15 mmol/L) or ionized calcium less than 4.65 mg/dL (1.16 mmol/L) with severe finding requiring inpatient management (eg, Altered mental status, seizures, arrhythmia, cardiac conduction disturbance)(10)
 - Calcium[C] greater than 14 mg/dL (3.5 mmol/L) or ionized calcium greater than 10.0 mg/dL (2.50 mmol/L) not corrected to near normal or near chronic baseline despite observation care treatment(11)
 - Calcium[C] greater than 12 mg/dL (3 mmol/L) or ionized calcium greater than 8.0 mg/dL (2.0 mmol/L) with severe finding requiring inpatient management (eg, Altered mental status, arrhythmia, cardiac conduction disturbance)(11)
 - Phosphorus less than 1 mg/dL (0.32 mmol/L) not corrected to near normal or near chronic baseline despite observation care treatment
 - Phosphorus less than 1.5 mg/dL (0.48 mmol/L) with severe finding requiring inpatient management (eg, Altered mental status, seizures)
 - Phosphorus greater than 10 mg/dL (3.2 mmol/L) not corrected to near normal or near chronic baseline despite observation care treatment
 - Phosphorus greater than 4.5 mg/dL (1.45 mmol/L) with severe finding requiring inpatient management (eg, crush injury, Altered mental status, seizures, arrhythmia, cardiac conduction disturbance)
 - Magnesium less than 1 mg/dL (0.41 mmol/L) not corrected to near normal or near chronic baseline despite observation care treatment
 - Magnesium less than 1.5 mg/dL (0.62 mmol/L) with severe finding requiring inpatient management (eg, Altered mental status, seizures, arrhythmia, cardiac conduction disturbance)
 - Magnesium greater than 4.9 mg/dL (2 mmol/L) not corrected to near normal or near chronic baseline despite observation care treatment
 - Magnesium greater than 3.0 mg/dL (1.23 mmol/L) with severe finding requiring inpatient management (eg, Altered mental status, arrhythmia, cardiac conduction disturbance)
 - Uric acid greater than 20 mg/dL (1.19 mmol/L) not corrected to near normal or near chronic baseline despite observation care treatment(12)
 - Uric acid greater than 8 mg/dL (0.48 mmol/L) with severe finding requiring inpatient management (eg, arrhythmia, cardiac conduction disturbance, seizure)(12)

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Footnotes

- A. In patients with hyperglycemia, a corrected serum sodium should inform decision making.
- B. ECG findings include peaked T-waves, PR interval greater than 0.20 seconds, QRS interval greater than 0.12 seconds, and arrhythmia.(2)(9)
- C. An ionized calcium level may be more accurate than a total serum calcium level, especially in the setting of acidosis, renal failure, multiple transfusions, hypoalbuminemia, or the presence of a paraprotein.(10)(11)

Severe renal failure

- Severe renal failure refers to renal failure or insufficiency necessitating inpatient dialysis or treatment with phosphate-binding agent medication (eg, sevelamer), or medication to treat secondary hyperparathyroidism (eg, cinacalcet).

Social Determinants of Health Assessment

- Risk of poor health outcomes may be increased by the presence of **1 or more** of the following social determinants of health(1)(2)(3)(4):
 - Housing insecurity, as indicated by **1 or more** of the following:
 - Individual or caregiver's current living situation is **1 or more** of the following(5):
 - Does not have own housing (eg, staying in a hotel, shelter, or with others)
 - Has own housing (eg, house, apartment), but at risk of losing it in the future (ie, behind on rent or mortgage)
 - Has own housing (eg, house, apartment), but has lived in 3 or more places in past year
 - Current housing has **1 or more** of the following:
 - Electrical appliances (eg, stove, refrigerator) not working or unavailable
 - Insufficient heating or cooling
 - Insufficient ventilation
 - Lead paint or pipes
 - Mold
 - Pests (eg, bugs) or rodents
 - Smoke detectors not working or unavailable
 - Food insecurity, as indicated by **1 or more** of the following(6):
 - In the past year, individual or caregiver ran out of food and did not have money to buy more food.
 - In the past year, individual or caregiver worried that they would run out of food before they received money to buy more food.
 - Insufficient transportation, as indicated by **1 or more** of the following(7):
 - In the past year, individual or caregiver missed medical appointments or could not get medications due to lack of transportation.
 - In the past year, individual or caregiver missed nonmedical activities, work, or could not get things needed for daily living due to lack of transportation.
 - Insufficient utilities, as indicated by **1 or more** of the following(8):
 - Utilities (eg, electricity, water, gas, or oil) are currently shut off or unavailable.
 - In the past year, electric, water, gas, or oil company threatened to shut off services.
 - Personal safety risk, as indicated by **2 or more** of the following(6):
 - Individual is sometimes or frequently physically hurt by another person (including family member).
 - Individual is sometimes or frequently insulted or talked down to by another person (including family member).
 - Individual is sometimes or frequently threatened with physical harm by another person (including family member).
 - Individual is sometimes or frequently screamed or cursed at by another person (including family member).

- Insufficient dependent care, as indicated by **1 or more** of the following:
 - In the past year, individual or caregiver was unable to work due to lack of dependent care.
 - In the past year, individual or caregiver was unable to work more (additional) hours due to lack of dependent care.
 - In the past year, individual or caregiver missed medical appointments or could not get medications due to lack of dependent care.
 - In the past year, individual or caregiver missed nonmedical activities (eg, school, church, social activity) due to lack of dependent care.
- Depression risk, as indicated by **ALL** of the following:
 - In the past 2 weeks, individual had little interest or pleasure in normal activities on at least several days.
 - In the past 2 weeks, individual felt down, depressed, or hopeless on at least several days.

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Tachycardia

- Tachycardia,[A][B] as indicated by **1 or more** of the following(1)(2):
 - Heart rate greater than 100 beats per minute in adult or child age 6 years or older
 - Heart rate greater than 115 beats per minute in child 3 to 5 years of age
 - Heart rate greater than 125 beats per minute in child 1 or 2 years of age
 - Heart rate greater than 130 beats per minute in infant 6 to 11 months of age
 - Heart rate greater than 150 beats per minute in infant 3 to 5 months of age
 - Heart rate greater than 160 beats per minute in infant 1 or 2 months of age

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Footnotes

- A. Criteria based upon clinician acquired numeric values (eg, vital signs, oxygen saturation) should be used if they are accurate reflections of the patient's condition. Transitory findings (eg, abnormal only upon initial emergency department intake or only one time out of multiple readings) that rapidly improve with no or minimal treatment usually do not reflect disease severity or risk for deterioration. This does not imply that an initial or one-time reading cannot ever be applicable. The goal is to separate erroneous or incidental findings from those that truly represent the patient's clinical picture.
- B. Interpretation of heart rate requires clinical judgment and consideration of several patient-specific factors, such as the patient's baseline heart rate, medications, and clinical impact. For example, an elderly patient on a beta-blocker medication with a baseline resting heart rate of 60 beats per minute may be clinically tachycardic at a heart rate of 94 beats per minute. Likewise, a patient who is upset, in pain, or nervous in the emergency department with a heart rate of 106 beats per minute may meet the technical definition of tachycardia, but this tachycardia (absent associated findings such as chest pain or hypotension) may not be clinically important. The numeric values included in this definition are provided to allow for consistency in terms of a technical definition of the

term tachycardia. Whether a heart rate above or below the technical threshold is clinically meaningful is a matter of persistence, context, and clinical judgment.

Tachycardia absent

- Tachycardia^{[A][B]} absent, as indicated by **1 or more** of the following(1)(2):
 - Heart rate less than or equal to 100 beats per minute in adult or child 6 years or older
 - Heart rate less than or equal to 115 beats per minute in child 3 to 5 years of age
 - Heart rate less than or equal to 125 beats per minute in child 1 or 2 years of age
 - Heart rate less than or equal to 130 beats per minute in infant 6 to 11 months of age
 - Heart rate less than or equal to 150 beats per minute in infant 3 to 5 months of age
 - Heart rate less than or equal to 160 beats per minute in infant 1 or 2 months of age

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Tachypnea

- Tachypnea^{[A][B]} as indicated by respiratory rate of **1 or more** of the following(2)(3):
 - Greater than 18 breaths per minute in adult or child 13 years of age or older^[A]
 - Greater than 22 breaths per minute in child 6 to 12 years of age^[A]
 - Greater than 25 breaths per minute in child 3 to 5 years of age^[A]
 - Greater than 30 breaths per minute in child 1 or 2 years of age^[A]
 - Greater than 40 breaths per minute in infant 6 to 11 months of age^[A]
 - Greater than 45 breaths per minute in infant 3 to 5 months of age^[A]
 - Greater than 60 breaths per minute in infant 1 or 2 months of age^[A]

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- B. Interpretation of respiratory rate requires clinical judgment and consideration of several patient-specific factors, such as the patient's baseline respiratory rate and whether the respiratory rate is indicative of significant underlying respiratory or other pathology. A mildly increased respiratory rate (eg, 20 to 22 in an adult), absent other indications of serious illness (eg, hypoxemia, metabolic acidosis, decreased tidal volume, pain), may be transitory and not clinically important. The numeric values included in this definition are provided to allow for consistency in terms of a technical definition of the term tachypnea. Whether a respiratory rate above the technical threshold is clinically meaningful is a matter of persistence, context, and clinical judgment. In addition, the measurement of the respiratory rate is subject to error. A medical textbook states that because there can be significant breath to breath variation in respiratory rate, the rate should be assessed for at least 30 seconds, preferably for a full minute.⁽¹⁾ This same textbook also states that new technologies designed to measure the respiratory rate have not proved to be clinically useful.⁽¹⁾

Tachypnea absent

- Tachypnea^{[A][B]} absent, as indicated by respiratory rate of **1 or more** of the following⁽²⁾⁽³⁾:
 - Less than or equal to 18 breaths per minute in adult or child 13 years of age or older
 - Less than or equal to 22 breaths per minute in child 6 to 12 years of age
 - Less than or equal to 25 breaths per minute in child 3 to 5 years of age
 - Less than or equal to 30 breaths per minute in child 1 or 2 years of age
 - Less than or equal to 40 breaths per minute in infant 6 to 11 months of age
 - Less than or equal to 45 breaths per minute in infant 3 to 5 months of age
 - Less than or equal to 60 breaths per minute in infant 1 or 2 months of age

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Troponin

- Troponin elevation alone does not indicate acute myocardial infarction; factors to consider include⁽¹⁾⁽²⁾:
 - Patients can have elevated levels of cardiac troponin due to other causes of myocardial injury, either acutely or chronically (eg, heart failure, myocarditis, pulmonary embolism, cardiomyopathy, defibrillator shock).
 - Patients with an elevated troponin should, on subsequent measurement, have a fall or rise in the troponin level of more than 20% to confirm acuity of myocardial injury. Absent this variation, the acuity of the myocardial injury should be questioned.
 - Attributing acute troponin release to myocardial ischemia depends on the presence of associated symptoms (eg, chest pain, dyspnea), ECG changes (eg, acute ST-segment deviation), and imaging findings (eg, new wall motion abnormalities on echocardiogram).
 - A report from a facility (eg, hospital) laboratory of an elevated troponin level, in general, indicates a level above the 99th percentile for the specific troponin assay used. The precise value that represents the 99th percentile depends on the specific troponin assay used.
 - The recommended time interval between serial troponin level assessments (eg, to detect fall or rise in troponin level) is 1 to 3 hours for high-sensitivity troponin, and 3 to 6 hours for conventional (contemporary) troponin assays.

- Evidence-based specialty society guidelines recommend use of high-sensitivity troponin assays, rather than older less-sensitive assays (collectively labeled contemporary assays).(1)(3) This is due to more rapid detection or exclusion of myocardial injury and increased diagnostic accuracy with high-sensitivity assays.(1)(3)

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Codes

ICD-10 Diagnosis: I09.81, I11.0, I13.0, I13.2, I50.1, I50.20, I50.21, I50.22, I50.23, I50.30, I50.31, I50.32, I50.33, I50.40, I50.41, I50.42, I50.43, I50.814, I50.82, I50.83, I50.84, I50.89, I50.9, T86.22 [Hide]

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